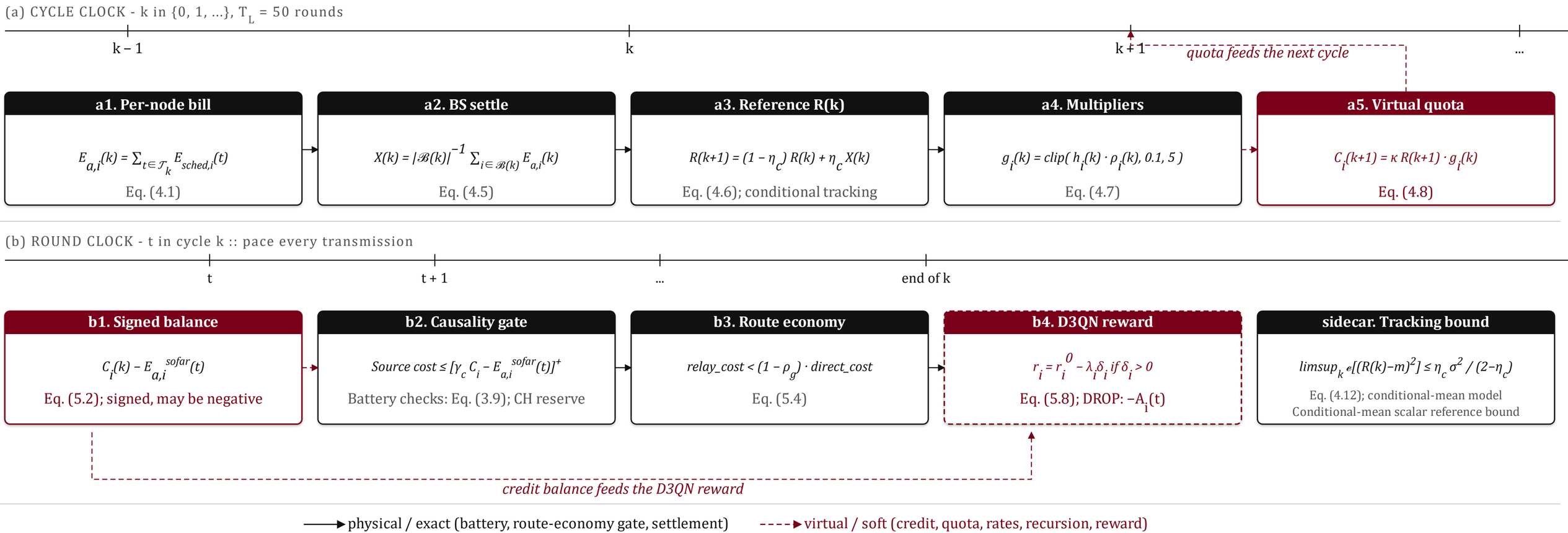

(a) CYCLE CLOCK - k in {0, 1, ...}, T_L = 50 rounds
k − 1
k
k + 1
...
quota feeds the next cycle
a1. Per-node bill
$E_{a,i}(k) = \sum_{t \in \mathcal{T}_k} E_{sched,i}(t)$
Eq. (4.1)
a2. BS settle
$X(k) = |\mathcal{B}(k)|^{-1} \sum_{i \in \mathcal{B}(k)} E_{a,i}(k)$
Eq. (4.5)
a3. Reference R(k)
$R(k+1) = (1 - \eta_c) R(k) + \eta_c X(k)$
Eq. (4.6); conditional tracking
a4. Multipliers
$g_i(k) = clip(\, h_i(k) \cdot \rho_i(k), 0.1, 5 \,)$
Eq. (4.7)
a5. Virtual quota
$C_i(k+1) = \kappa\, R(k+1) \cdot g_i(k)$
Eq. (4.8)
(b) ROUND CLOCK - t in cycle k :: pace every transmission
t
t + 1
...
end of k
b1. Signed balance
$C_i(k) - E_{a,i}^{sofar}(t)$
Eq. (5.2); signed, may be negative
b2. Causality gate
$Source\ cost \le [\gamma_c C_i - E_{a,i}^{sofar}(t)]^+$
Battery checks: Eq. (3.9); CH reserve
b3. Route economy
$relay_cost < (1 - \rho_g) \cdot direct_cost$
Eq. (5.4)
b4. D3QN reward
$r_i = r_i^0 - \lambda_i \delta_i \ if\ \delta_i > 0$
Eq. (5.8); DROP: $-A_i(t)$
sidecar. Tracking bound
$limsup_k\, \mathbb{E}[(R(k)-m)^2] \le \eta_c \sigma^2 / (2-\eta_c)$
Eq. (4.12); conditional-mean model
Conditional-mean scalar reference bound
credit balance feeds the D3QN reward
physical / exact (battery, route-economy gate, settlement)
virtual / soft (credit, quota, rates, recursion, reward)